

Polly — Pre-MVP3 Research: Executive Summary

Prediction-market (Polymarket) trading strategy & data program · compiled and primary-source-verified 2026-06-25 · one-page brief

The thesis (verified, unchanged): Detecting informed/insider activity on Polymarket is easy; **capturing** it is the hard part, and **following a confirmed fill is a losing race**. Permanent price impact lands in seconds (Kyle 1985), the public feed reveals trade direction only ~59% of the time (Dubach 2026), bots fill in milliseconds, and takers pay a category fee on a $p(1-p)$ curve. MVP3 therefore stops copying winners and instead **builds the data asset** and trades signals that **don't self-erase**, proven by survivorship-free forward paper-trading.

~3%
of traders drive price discovery (Yale/LBS, 1.72M accts, \$13.76B)

~0.64
news-passthrough → multi-minute drift (Kalshi study)

\$40M
internal arbitrage extracted in 1 yr — but ~15-share capacity

69.9%
win rate of the flagged insider cohort (Mitts & Ofir)

THE FOUR DEFENSIBLE EDGES

- 1. Passive market-making** VERIFIED — Polymarket subsidizes makers (no maker fee + 20–25% rebate + quadratic liquidity rewards). Most structurally robust; underwriting risk, so pull quotes around event conclusion.
- 2. Mispricing / fair value** — de-vig & ensemble external anchors; bet deviations. Strong for sports (Shin de-vig) and long-horizon politics (compression correction: 70¢ a month out ≈ ~75% true). Internal arb is real but tiny-capacity.
- 3. Fade the dumb money** VERIFIED — ~3% skilled minority captures >30% of gains; the unskilled majority funds them. Large, persistent, **not self-erasing** when observed.
- 4. News-latency** — react to news faster than slow humans in the ~0.64-passthrough window. The one genuine "fast" race; expect MEV/HFT competition.

INSIDER DETECTION → USED DEFENSIVELY

- Not a copy signal (the fill *is* the signal). Use as a **toxicity filter** for market-making and a **fade confirmation**.
- Composite score only — fresh-wallet + shared-CEX funding cluster + concentration + last-minute taking + anomalous win-rate — to avoid "Théo" false positives.

WHAT MVP3 BUILDS

- Always-on collector first** — full L2 order books + trade tape + on-chain flows + funding graph + resolutions + off-chain anchors, **bitemporal**, with the market universe fixed at t_0 (kills survivorship).
- Point-in-time firewall** — decision layer may only call `as_of(t)` (the look-ahead defense mvp1/mvp2 lacked).
- Strategies in robustness order:** market-making → mispricing → fade-dumb-money → news-latency → insider (defensive).
- Forward paper-trade**, evaluated at **1 / 2 / 6 months** with anytime-valid stats; report win-rate first, deflate the Sharpe, state MinTRL honestly.

KEY VERIFIED FACTS THAT SHAPE THE BUILD

Item	Verified value
v2 OrderFilled	NEW topic0 + separate FeeCharged → dual decoder
UMA liveness	2h fast path (48h is DVM-only)
Feed direction	~59% accurate → signed flow from on-chain only
538 polls	shut down Mar 2025 → dropped as an anchor
Forward-test stats	PSR/MinTRL/DSR, Lo-SE, e-values byte-verified

VERIFICATION VERDICT

Re-fetched every load-bearing claim against primary sources. **Strategic conclusions stand**. Corrections were to numbers/citations only. The one reversal — the favorite-longshot bias is **classic, not reversed** CORRECTED — *reinforces* the fair-value edge.

Sources: `research/PRE_MVP3_RESEARCH.md`, `research/MVP3_DESIGN.md`, `research/VERIFICATION_LOG.md`. Primary-source verified where the publisher allowed automated fetch (arXiv, docs.polymarket.com, Polygonscan, reference code); a minority remain cross-checked search snippets where hosts (SSRN, ResearchGate, Yale Insights) return origin anti-bot blocks.